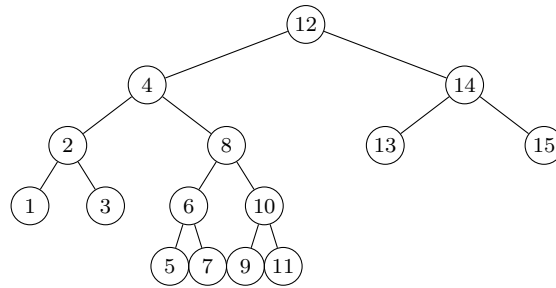


CSCI-UA 102 — Quiz 9

AVL Trees / Rotations

Name: _____ NetID: _____

Consider the following binary search tree containing the keys 1 through 15:



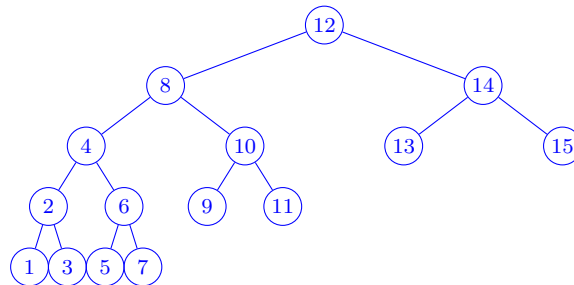
(5 points) Suggest a series of rotations that turns this into a balanced tree. Draw the BST after each rotation.

Solution

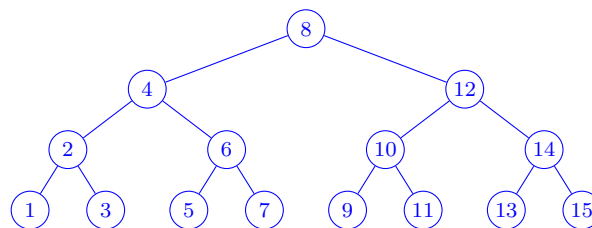
The tree is unbalanced at the root: the left subtree (rooted at 4) has height 4, the right subtree (rooted at 14) has height 2. Balance factor at 12 is -2 .

The left child 4 is itself **right-heavy**: its left subtree (rooted at 2) has height 2, while its right subtree (rooted at 8) has height 3. So this is a **left-right (LR) case** at node 12, requiring two rotations.

Step 1: Left rotation at 4. Promote 8 in place of 4; 4 becomes 8's left child; 8's old left child (6) becomes 4's new right child.



Step 2: Right rotation at 12. Promote 8 in place of 12; 12 becomes 8's right child; 8's old right child (10) becomes 12's new left child.



The tree is now perfectly balanced (height 3, all leaves at the same level). The two rotations together form a **left-right rotation** at node 12.